

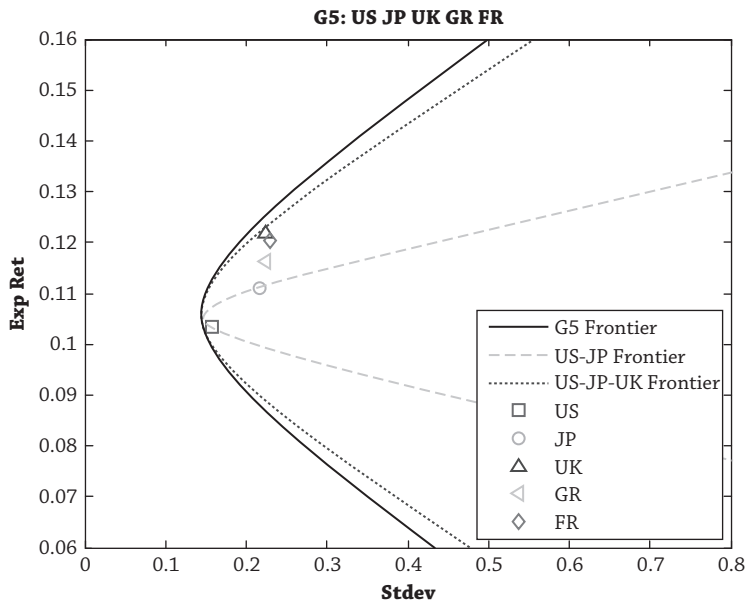


Figure 3.5

dashed line. Two things have happened in moving from the G2 (United States and Japan) to the G3 (United States, Japan, and the United Kingdom):

1. The frontier has expanded.

The Happy Buddha becomes much happier adding the United Kingdom. The pronounced outward shift of the wings of the mean-variance bullet means that an investor can obtain a much higher return for a given standard deviation. (There is also a leftward shift of the frontier, but this is imperceptible in the graph.) Starting at any point on the United States–Japan frontier, we can move upwards on an imaginary vertical line and obtain higher returns for the same level of risk. Adding the United Kingdom to our portfolio provides further diversification benefits because now there is an additional country that could have high returns when the United States is in a bear market while before there was only Japan. There is also a chance that both the old United States and Japan positions would do badly; adding the United Kingdom gives the portfolio a chance to offset some or all of those losses.

2. All individual assets lie inside the frontier.

Individual assets are *dominated*: diversified portfolios on the frontier do better than assets held individually. Now all countries would never be held individually. Diversification removes asset-specific risk and reduces the overall risk of the portfolio.

In Figure 3.5, we add Germany and France. The G5 mean-variance frontier is in the solid line, and it is the fattest: the United States–Japan and the United